*Week 7, July-Sept 2017*
*MOOCS Decision Modeling Course*

# Networks Models

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Module 31

# Introduction to Graph Theory

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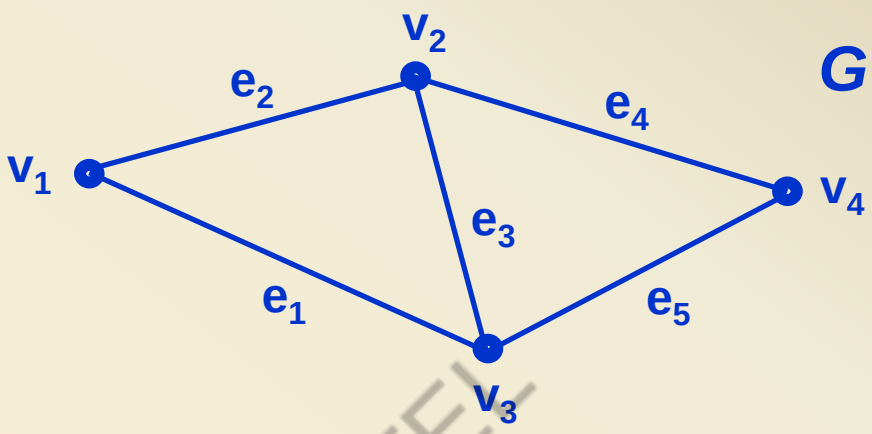
# What is a Graph

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**A Graph G is a set of vertices and edges**

**Null graph**

**A graph with no vertices and edges**

# Graph Theory Examples

- Utilities problem
- Electrical network problem
- Seating problem
- Flow problem
- Allocation problem
- Transportation problem
- Social networks

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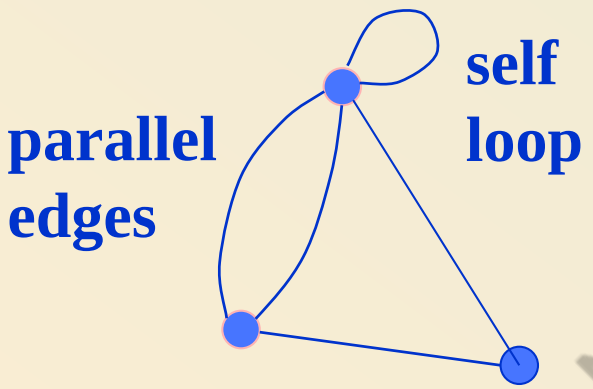
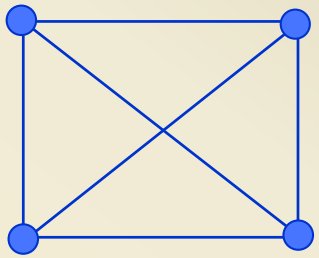
# Simple graph

- **Graphs may have self loops and parallel edges – A simple graph does not have them.**




**Not a simple graph**



**A simple graph**

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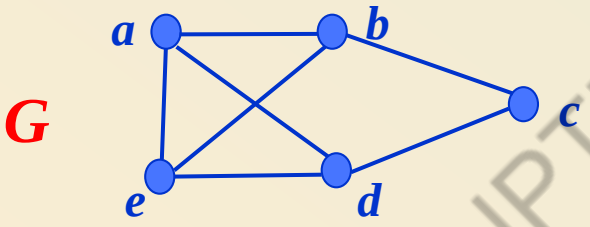
# Subgraphs

- A **sub graph** is a subset of a graph

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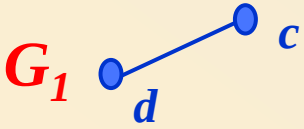


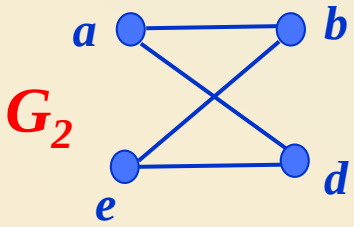


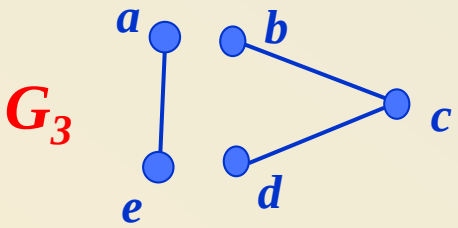

# Incidence and Adjacency

- Edge $e_i$ has vertices $(v_j, v_k)$
- $v_j$ and $v_k$ are then adjacent
- Edge $e_i$ is incident upon $v_j$

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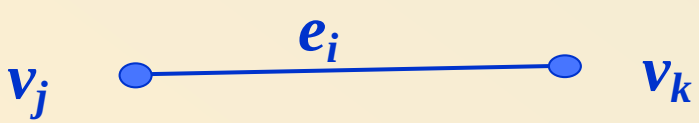
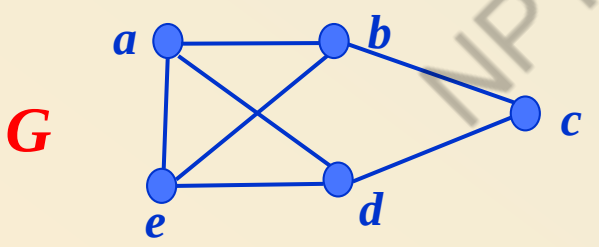


# Degree of a Vertex

**Degree of vertex *v* in graph *G*, *d* (*v*), is the number of edges incident on *v*, with each self-loop counted twice**

- **The maximal degree is $\Delta(G)$**
- **The minimum degree is $\delta(G)$**




- **What is the degree of vertex a?**

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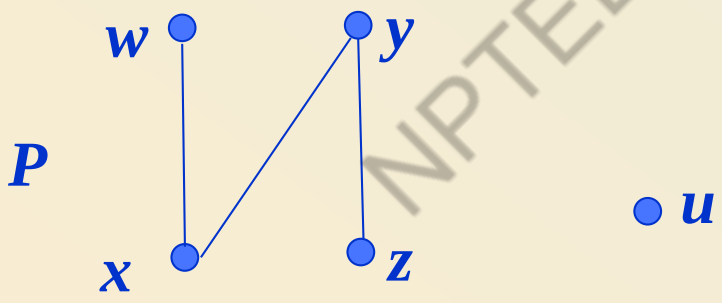
# Isolated and pendant Vertices

- In Graph P, u is an isolated vertex.
- In Graph P, w and z are pendant vertices

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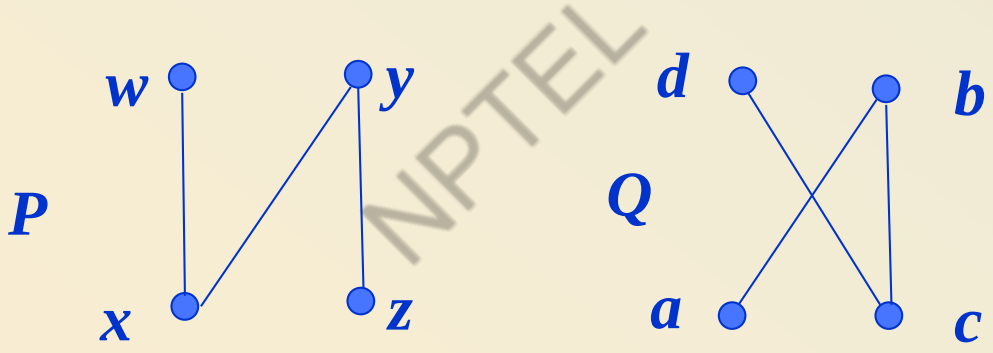
# Isomorphism

- **Isomorphic graphs have one to one incidence relationship between their vertices and edges.**

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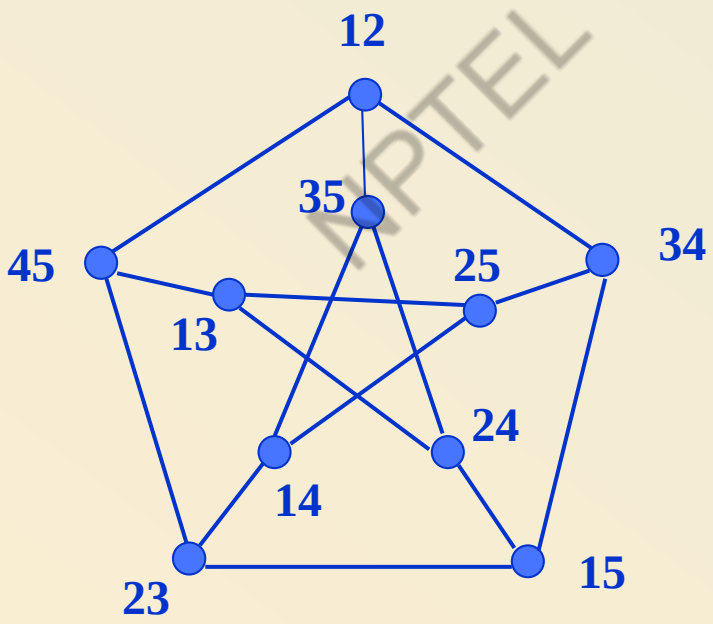
# Petersen Graph

- The **Petersen Graph** is the simple graph whose vertices are the 2-element subsets of a 5-element set and whose edges are pairs of disjoint 2-element subsets

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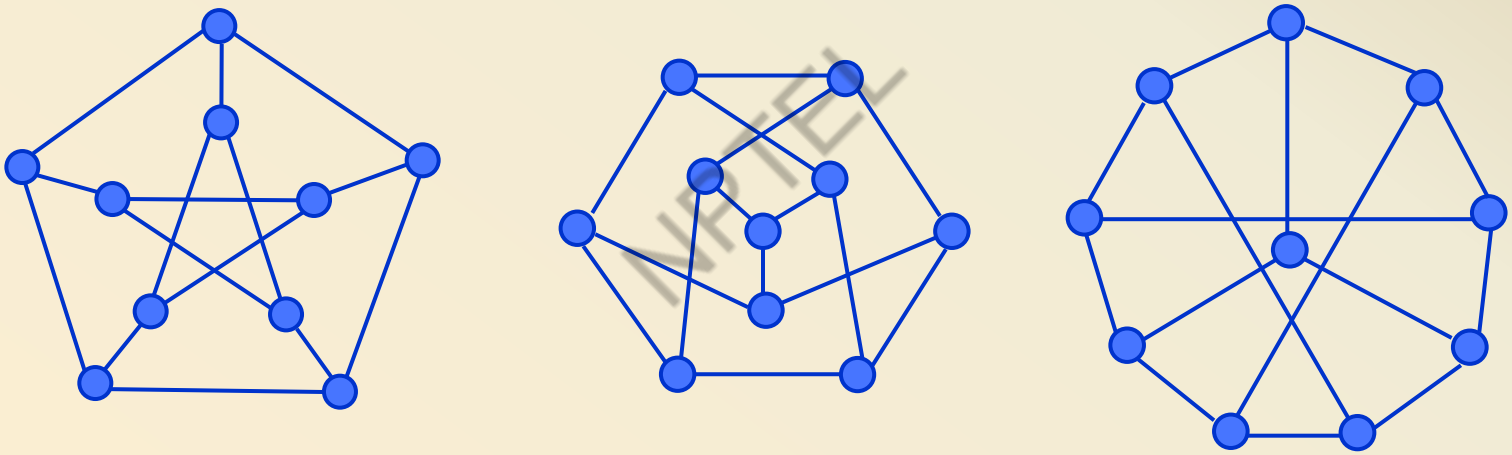
# Petersen Graph

**3 isomorphic forms**



# Edge and Vertex Disjoint Subgraphs

- **Edge disjoint sub graphs – sub graphs of a Graph G with no common edge between them.**

- **Vertex disjoint sub graphs – sub graphs of a Graph G with no common vertex between them.**

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# Introduction (Continued)

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# Walk

**A walk is a finite alternate sequence of vertices and edges beginning and ending in vertices with each edge incident on vertices following and preceding it.**

- **Open u-v walk** $(u \neq v)$ $v_1$-$e_2$-$v_2$-$e_3$-$v_3$-$e_5$-$v_4$-$e_4$-$v_2$-$e_3$-$v_3$
- **Closed u-v walk** $(u = v)$ $v_2$-$e_3$-$v_3$-$e_5$-$v_4$-$e_4$-$v_2$-$e_3$-$v_3$-$e_3$-$v_2$

***Note: Edges and vertices may be repeated***

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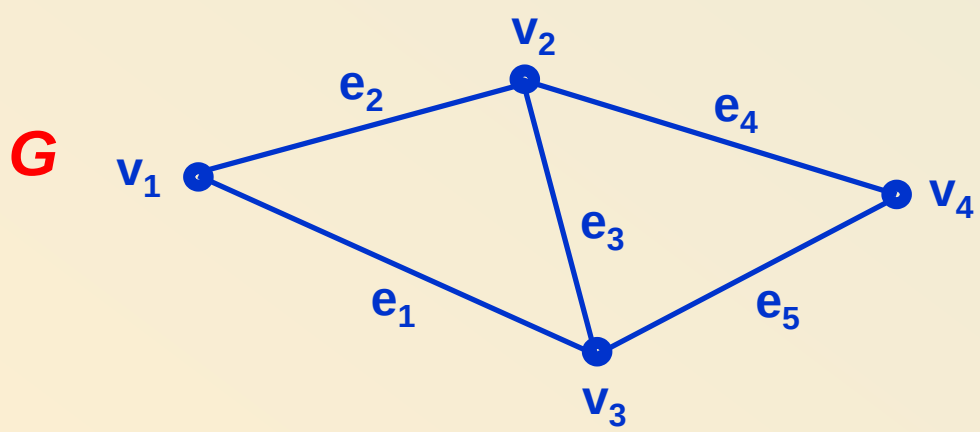

# Trail

**A Trail is a walk with no edge repeated**

- **Open u-v trail** $(u \neq v)$ $v_1$-$e_2$-$v_2$-$e_3$-$v_3$-$e_5$-$v_4$-$e_4$-$v_2$
- **Closed u-v trail** $(u = v)$ $v_2$-$e_3$-$v_3$-$e_5$-$v_4$-$e_4$-$v_2$

***Note: vertices may be repeated***

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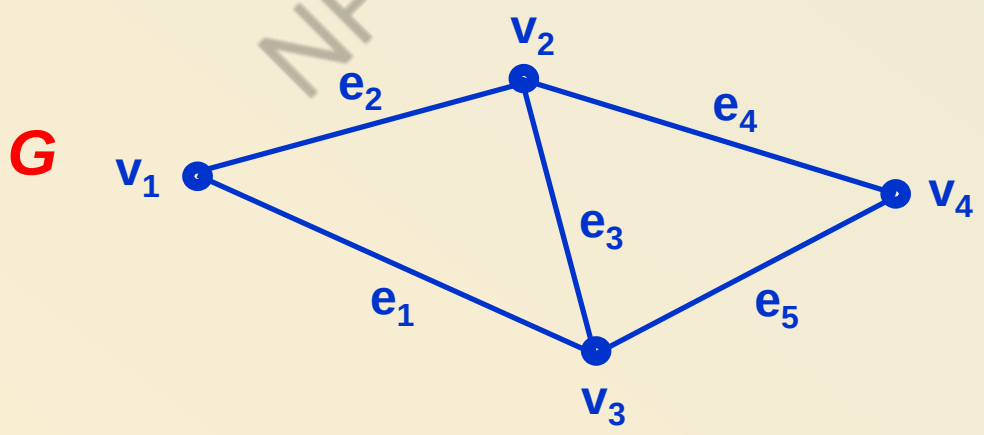
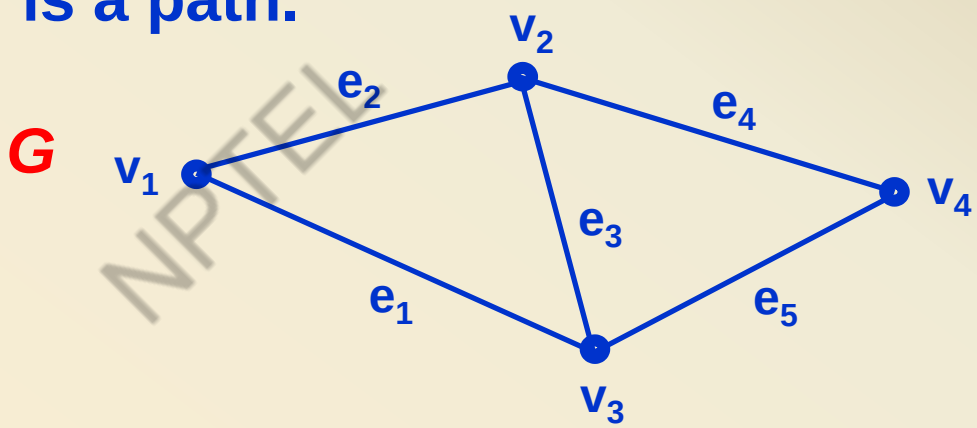


# Path

A **Path** is a **trail** with **no vertex repeated** – a sequence of **distinct** vertices such that two consecutive vertices are adjacent.

- $v_1$-$e_2$-$v_2$-$e_3$-$v_3$-$e_5$-$v_4$ is a path.

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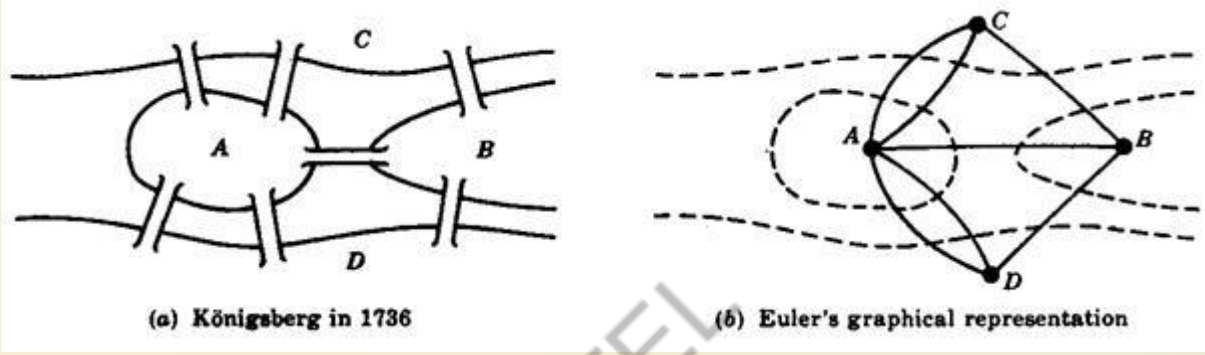
A **closed path** is a cycle or a **circuit** (*note: beginning and end vertices are same – no other vertices can be repeated*)

$v_2$-$e_3$-$v_3$-$e_5$-$v_4$-$e_4$-$v_2$ is a **circuit** but

$v_2$-$e_3$-$v_3$-$e_3$-$v_2$ is a **closed walk**.

# The Königsberg Bridge Problem




(a) Königsberg in 1736 (b) Euler's graphical representation

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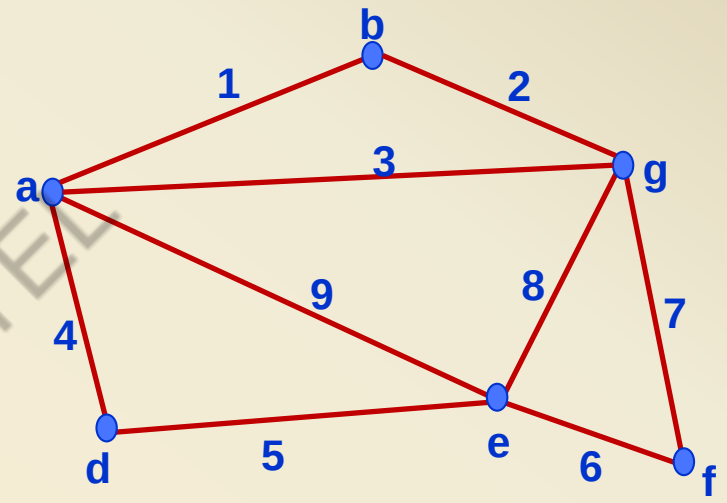
- There are four landmasses and seven bridges.
- How to cross all the bridges exactly once starting from any of the four landmasses and coming back to the same landmass?

# Euler graph

- In Euler graphs, it is possible to traverse all the edges exactly once starting from any of the vertices and coming back to the same vertex.
- Is the graph shown here an Euler graph?
- In a Euler Graph, all the vertices are of even degree

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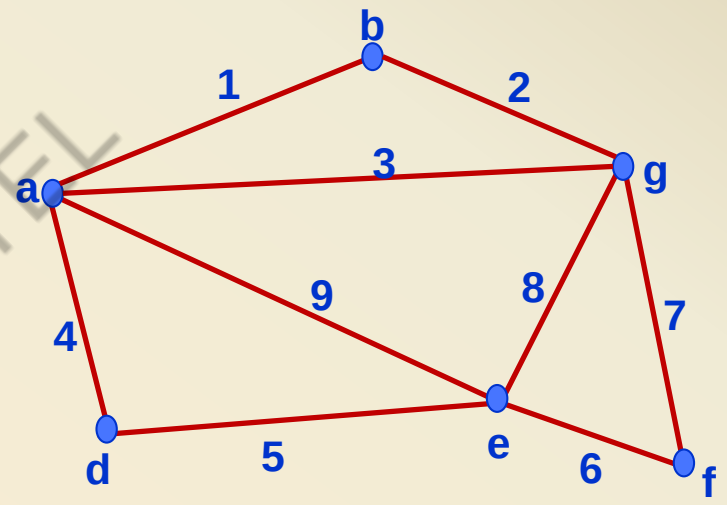
- An Euler Line or Euler Trail crosses every edge exactly once without repeating, and ends at the initial vertex.

# Euler graph

- In Euler graphs, it is possible to traverse all the edges exactly once starting from any of the vertices and coming back to the same vertex.

- Edge Sequence
  1-2-3-4-5-6-7-8-9 or
  Node Sequence
  a-b-g-a-d-e-f-g-e-a
  is an Euler Trail.

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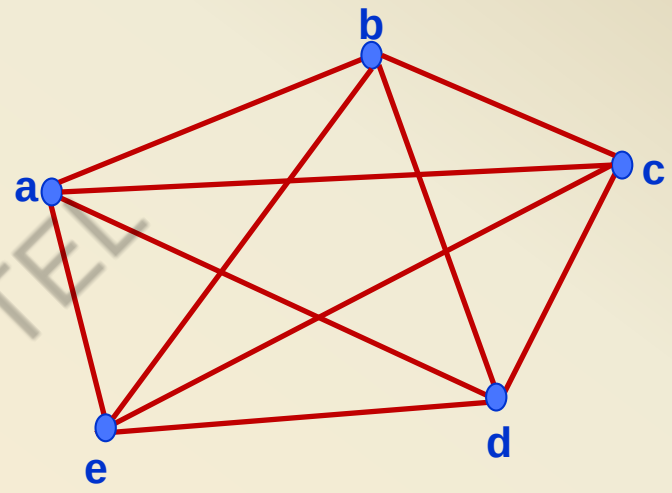
- Euler trail is a
  a Closed trail covering
  all the edges exactly once.

- In a Chinese Postman problem, the solution is a Euler trail.

# Hamiltonian Circuit

- A **Hamiltonian Circuit** in a graph is a circuit that covers **every vertex exactly once** beginning and ending at the same vertex.

- Node Sequence
  a-b-c-d-e is a
  **Hamiltonian Circuit**.

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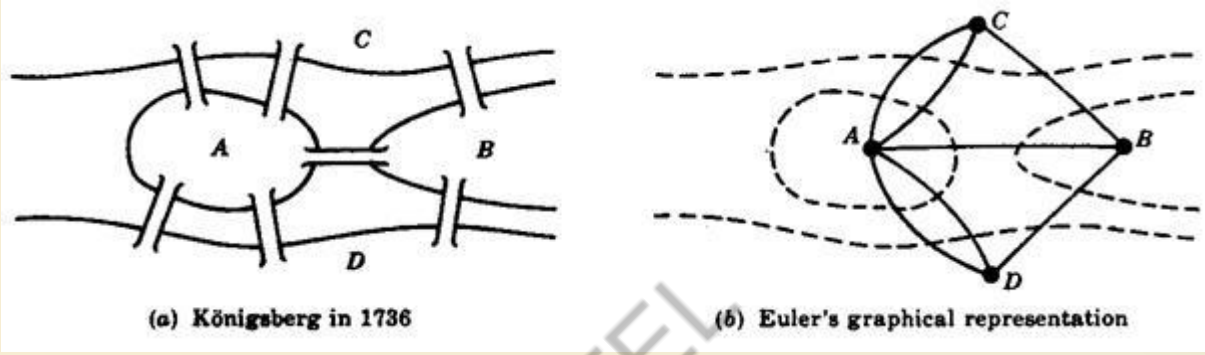
- Are there **more** such circuits?

- What about a-e-b-d-c-a? or a-c-b-e-c-d-a?

- There could be other Hamiltonian circuits as well.

- In a **Travelling Salesman** problem, the solution is a **Hamiltonian circuit**.

# The Königsberg Bridge Problem




(a) Königsberg in 1736 (b) Euler's graphical representation

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- Is the Graph representing the problem an Euler Graph?
- What are the degrees of the vertices?
- Are they all even?

Königsberg Bridge Problem requires the finding of a closed trail.

# Arbitrarily Traceable Euler Graphs

- Arbitrarily Traceable Euler graphs are Euler graphs where it is possible to traverse all the edges exactly once starting from any of the vertices and coming back to the same vertex by arbitrarily choosing vertices all the time.
- Are the following graphs arbitrarily traceable Euler graphs?

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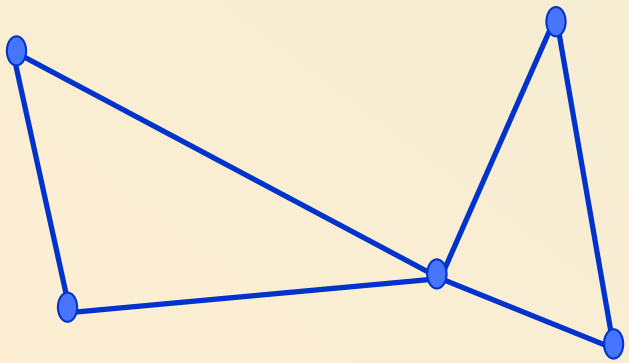
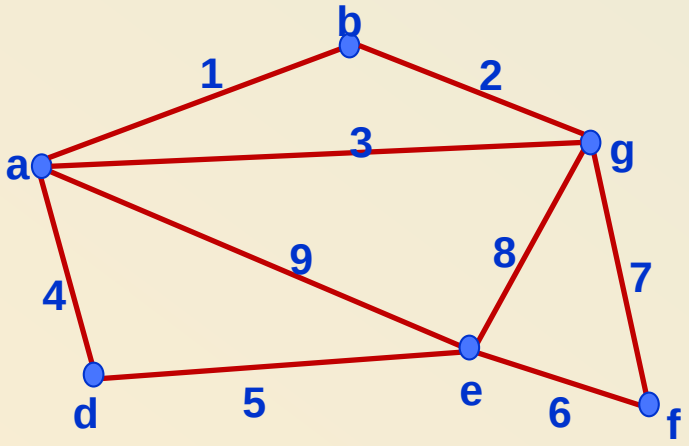
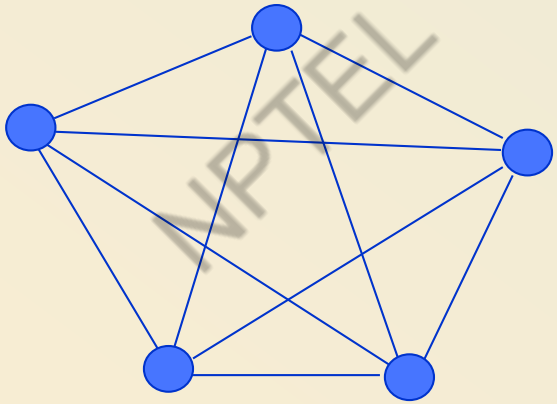




# Complete Graph

- **A complete graph is a simple graph with edges between every pair of vertices**



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# Connected and Disconnected Graphs

- ***Connected*** **: There exists at least one path between two vertices**
- ***Disconnected*** **: Otherwise**
- **Example:**
  - $G_1$ **and** $G_2$ **are connected**
  - $G_3$ **is disconnected**

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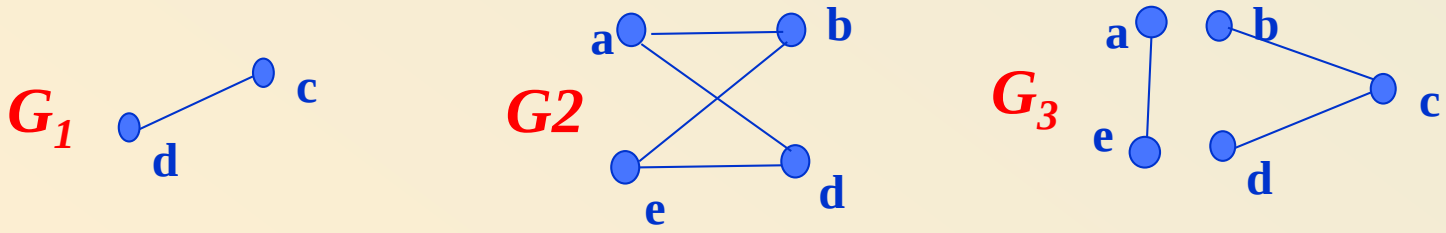

# Order and Size of a Graph

**n(G): Order of a graph G – is the number of vertices in G.**

**An n-vertex graph is a graph of order n.**

**e(G): Size of a graph G – is the number of edges in G.**

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Module 33

# Operations on a Graph Tree and Spanning Tree

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# Operations on a graph

**Union**: Union of edges and vertices

$$G(v, e) = G1(v_1, e_1) \cup G2(v_2, e_2)$$

$$v = v_1 \cup v_2 \quad ; \quad e = e_1 \cup e_2$$

**Intersection**: Common vertices and edges

$$G(v, e) = G1(v_1, e_1) \cap G2(v_2, e_2)$$

$$v = v_1 \cap v_2 \quad ; \quad e = e_1 \cap e_2$$

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# Operations on a graph

**Ring sum**

- **Union of vertices and edges in G1 or G2 minus those vertices and edges that are in both**
- **Union minus Intersection**

$$G(v, e) = G1(v_1, e_1) \oplus G2(v_2, e_2)$$

$$v = v_1 \oplus v_2 = (v_1 \cup v_2)$$

$$e = e_1 \oplus e_2 = (e_1 \cup e_2) - (e_1 \cap e_2)$$

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# Operations on a graph

## Decomposition

- G is said to be decomposed into G1 and G2 if:
- G = G1 U G2 and G1 ∩ G2 is a null graph

$$G(v, e) = G1(v_1, e_1) \cup G2(v_2, e_2)$$

and $G1(v_1, e_1) \cap G2(v_2, e_2)$ is a null graph $\Phi$

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# Operations on a graph

## Deletion

- Vertex deletion deletes all edges incident on it
- Edge deletion does not delete vertices

## Fusion

- Vertices are fused
- With fusion, no of vertices reduces by 1
- No of edges remains same – edges between the two fused vertices become self-loops

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# Tree

**A tree is a:**

- **A connected graph without circuit**
- **A minimally connected graph**
- **A connected graph with n vertices and n-1 edges**

**In a tree, there is only one path between any pair of vertices**

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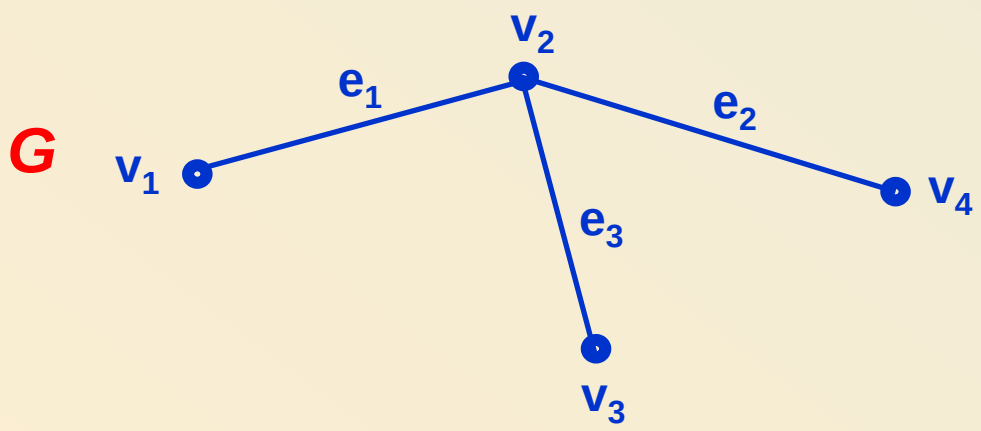
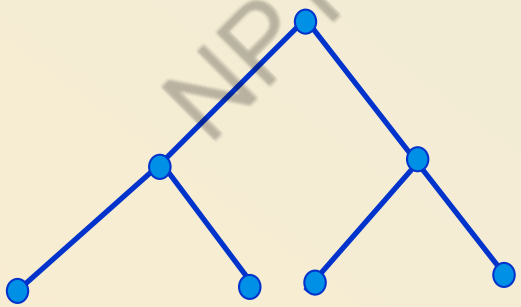


# Binary Tree

**Binary tree**

- **Exactly one vertex of degree 2 - others of 1 or 3**
- **No of vertices in a binary tree is odd**



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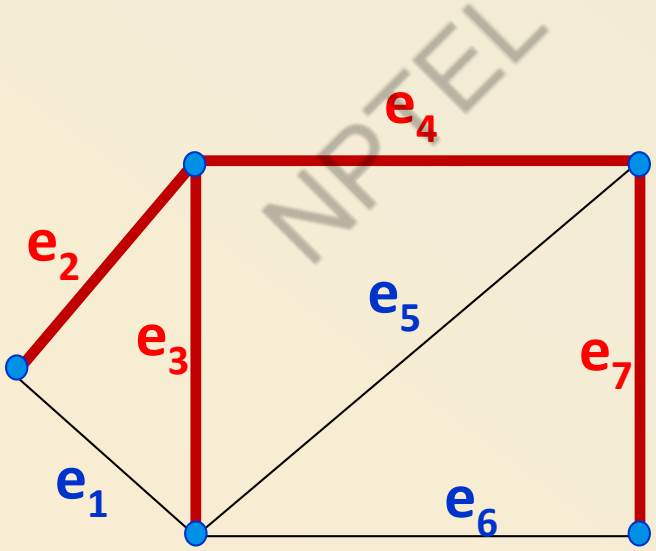
# Spanning Tree

**Spanning tree**

- **A tree T that is a sub graph of connected graph G and contains all the vertices of G**
- **A spanning tree is indicated in the graph below**

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# Spanning Tree

**Spanning tree**

- Can there be a spanning tree for a disconnected graph?
- Does every connected graph have a spanning tree?
- Are spanning trees unique in each graph?

A disconnected graph of k components has a spanning forest consisting of k spanning trees of k corresponding components

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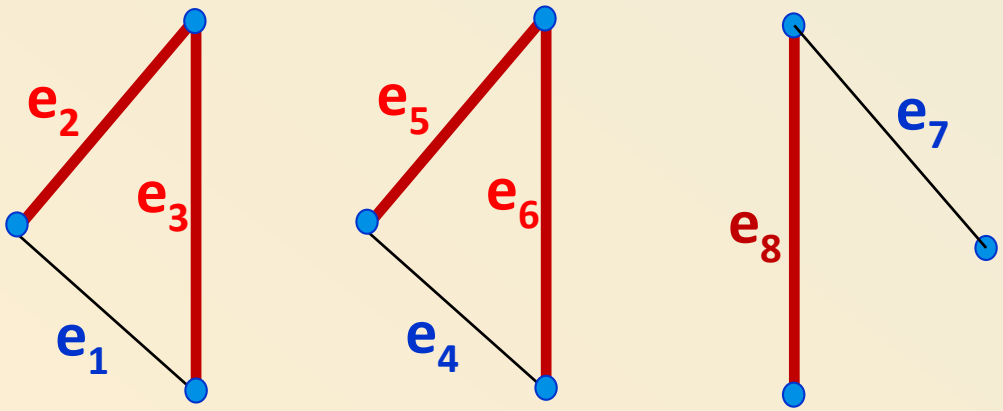
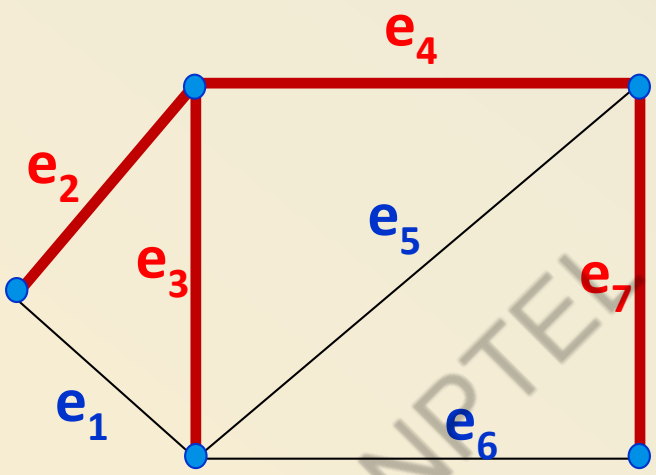


# Spanning Tree

- How many different spanning trees the following graph contain?

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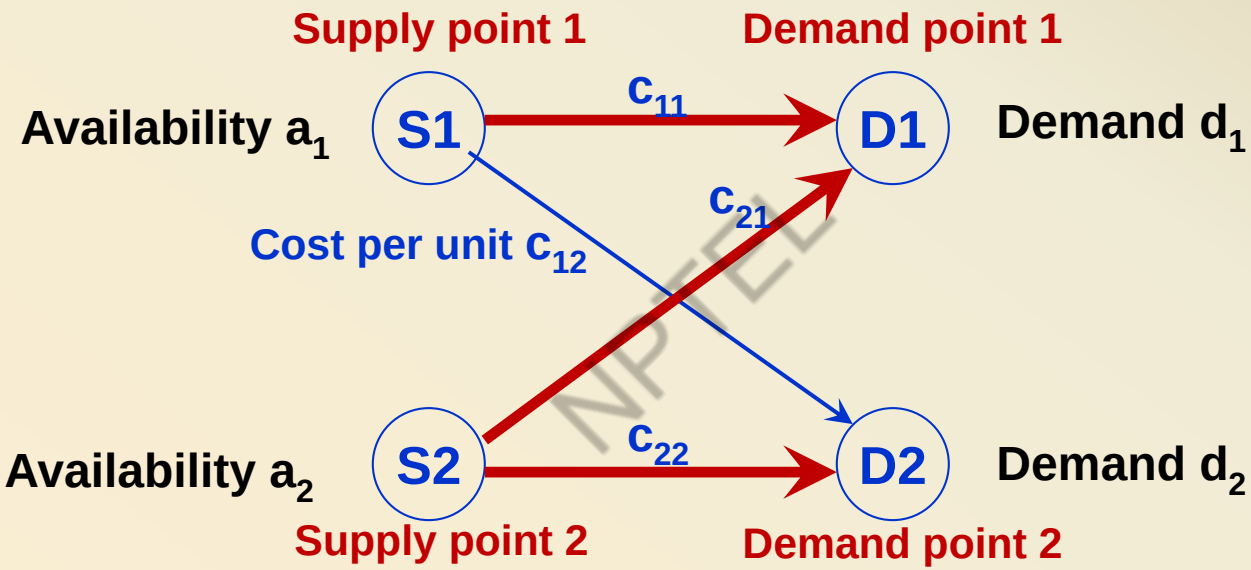
$S_1 = \{e_2 , e_3 , e_4 , e_7\}$ $S_2 = \{e_1 , e_2 , e_4 , e_7\}$
$S_3 = \{e_1 , e_3 , e_4 , e_7\}$ $S_4 = \{e_1 , e_4 , e_5 , e_7\}$
etc. etc.

What do these different spanning trees signify?

# LP and Spanning Tree

**Consider the LP Problem – Find optimal solution for the transportation problem.**

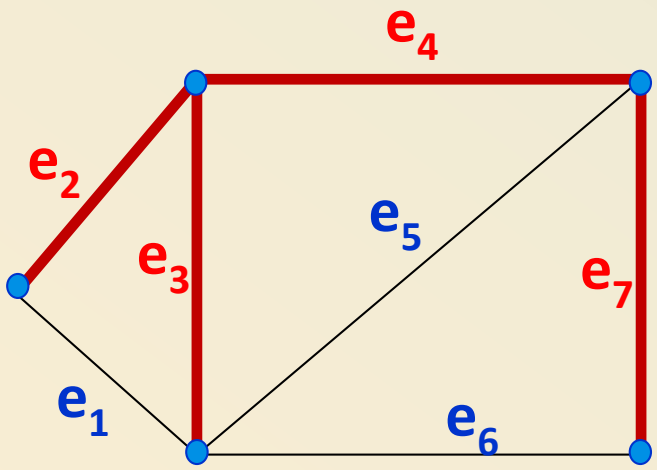



- **Each spanning tree of the graph representing the transportation problem is a candidate solution or a basic solution in the LP language.**
- **One Spanning tree {S1-D1, S2-D1, S2-D2} is shown in red.**

# Chords

**Chords, ties or links**

- **All the remaining edges of a graph G other than the spanning tree edges**
- **With respect to a spanning tree T [T = {$e_2$ , $e_3$ , $e_4$ , $e_7$}], The chords, ties, or links are $e_1$ , $e_5$ , $e_6$**
- **What happens with another spanning tree?**

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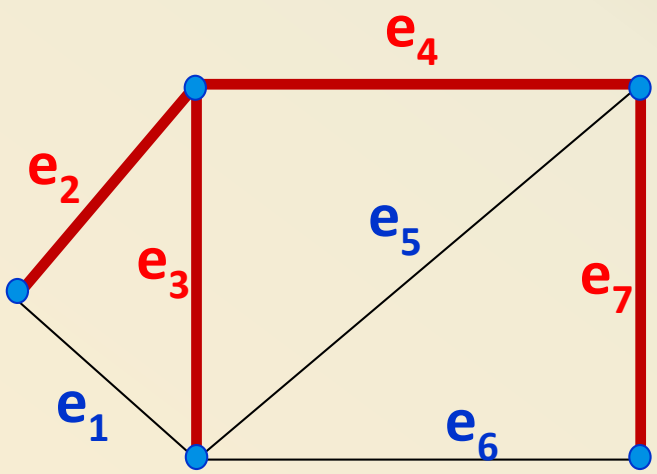
# Chords

- With respect to a spanning tree T, each chord gives rise to exactly one circuit with the spanning tree edges - such a circuit is called a fundamental circuit.

- With respect to a spanning tree T [T = {$e_2$ , $e_3$ , $e_4$ , $e_7$}], The chords, ties, or links are $e_1$ , $e_5$ , $e_6$

- Can you identify the Fundamental Circuits?

- How will you find other circuits in the graph?

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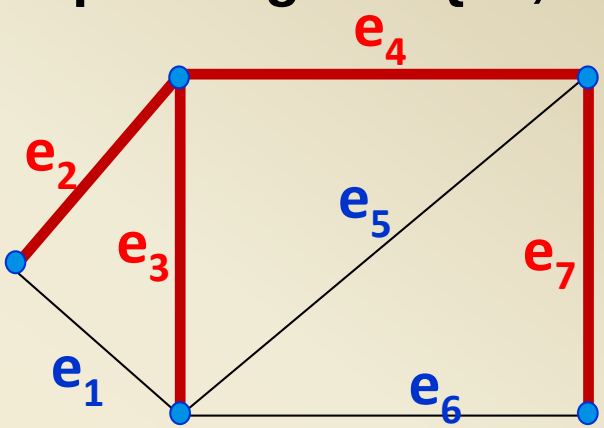
# Fundamental Circuit Matrix

**Consider the graph. With respect to the spanning tree {e2, e3, e4, e7}, find out the following:**




- **Fundamental Circuit matrix ($B_f$)**

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## Fundamental Circuit matrix ($B_f$)

| | $e_2$ | $e_3$ | $e_4$ | $e_7$ | $e_1$ | $e_5$ | $e_6$ |
|---|---|---|---|---|---|---|---|
| $B_1$ | 1 | 1 | 0 | 0 | 1 | 0 | 0 |
| $B_2$ | 0 | 1 | 1 | 0 | 0 | 1 | 0 |
| $B_3$ | 0 | 1 | 1 | 1 | 0 | 0 | 1 |

Module 34

# Minimal Spanning Tree

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# Minimal spanning tree

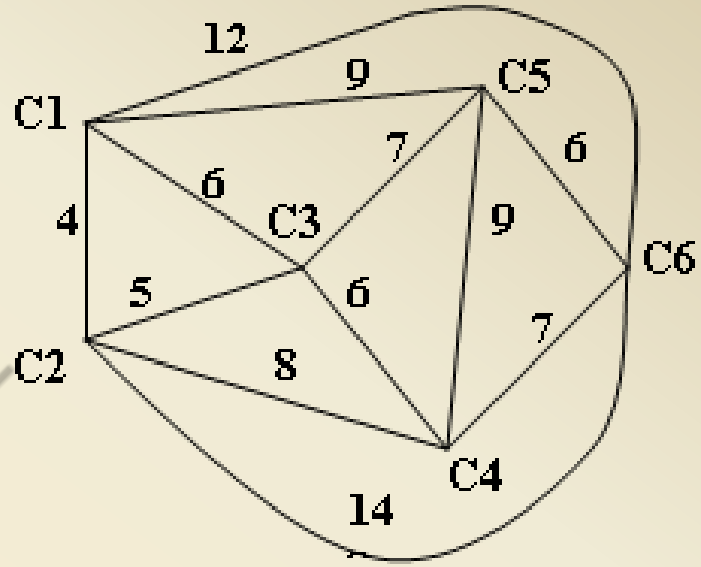
Consider this graph.

It has weights associated with the edges.

What do they signify?

This could be a road network

With weights denoting distance




This could be a social network with weights denoting strength of correlation.

Then what a minimal connection between the nodes would signify?

What about maximal connection between the nodes?

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# Minimal spanning tree

**Minimal Connection between the nodes can be found out in a graph by finding minimal spanning tree of the graph.**

**Two algorithms are popular:**

**1) Kruskal's Algorithm**

- **Edge-based**
- **rely on finding next lowest edges that do not make circuits with the previously selected ones**
- **implementation is difficult in computers**

**2) Prim's Algorithm**

- **Node-based**
- **rely on connecting a node with its nearest neighbor.**
- **Implementation is easier**

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# Minimal spanning tree

**Kruskal's algorithm**

- **The Algorithm is edge based**

**Steps**

1) ***Sort all the edges of the Graph with n vertices as per their ascending weights***

2) ***Start with the edge of minimum weight***

3) ***Consider another edge of next higher weight edges***

4) ***Ensure that the edge on consideration do not make any circuit with the already selected ones***

5) ***Repeat the steps 3) and 4) till n-1 edges are selected.***

6) ***The selected edges forms a minimal spanning tree.***

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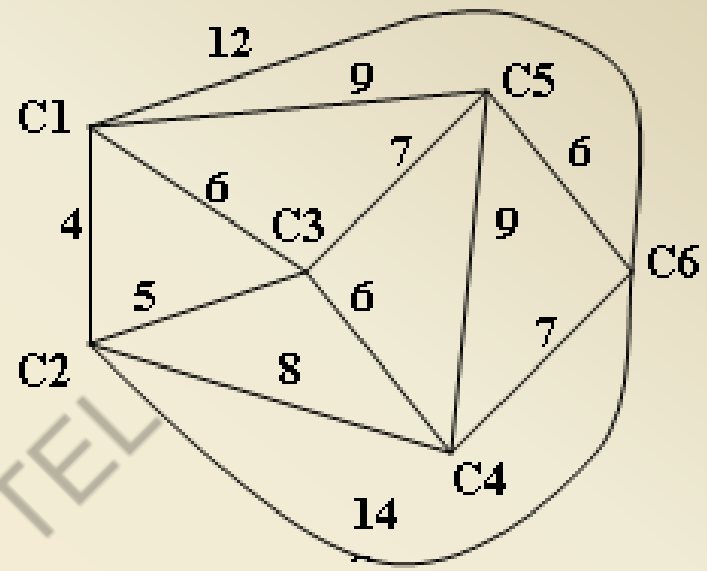
# Minimal spanning tree

**Kruskal's algorithm**

**Find Minimal Spanning tree (MST) for the graph shown:**

**How do we proceed to obtain MST Using Kruskal's Algorithm?**

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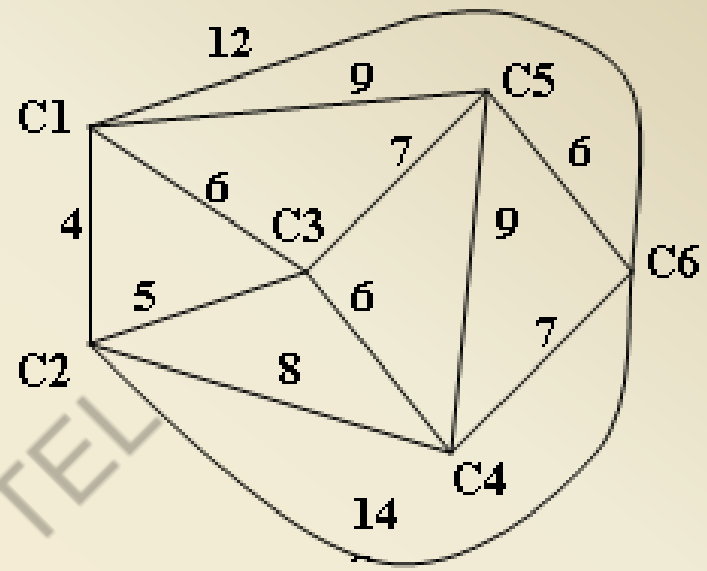
- ***First, we sort the edges. C1-C2 is the minimum weight edge. Next we consider C2-C3 – can we choose this one?***
- ***Next we consider C1-C3 – does it make a circuit?***
- ***Next Consider C3-C4 – can it be selected?***
- ***Next Consider C5-C6 – can we choose this one?***
- ***Now 4 edges are selected – which one is 5$^{th}$ or the last? (remember there are 6 vertices) – is it C3-C5 or C4-C6?***

# Minimal spanning tree

**Kruskal's algorithm**

**Find Minimal Spanning tree (MST) for the graph shown:**




**Two Solutions emerge – both with a total weight of 28.**

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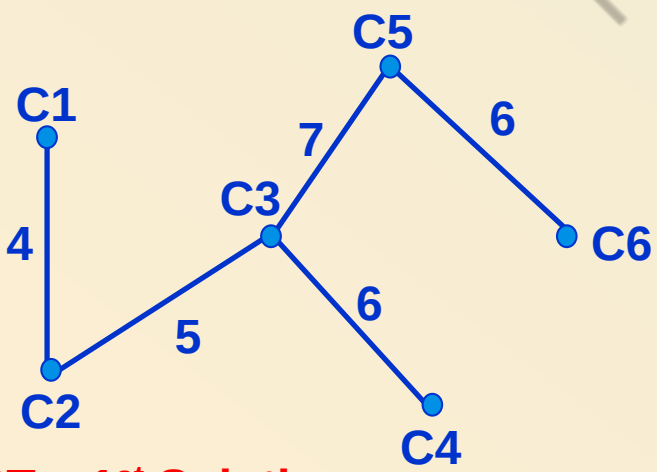
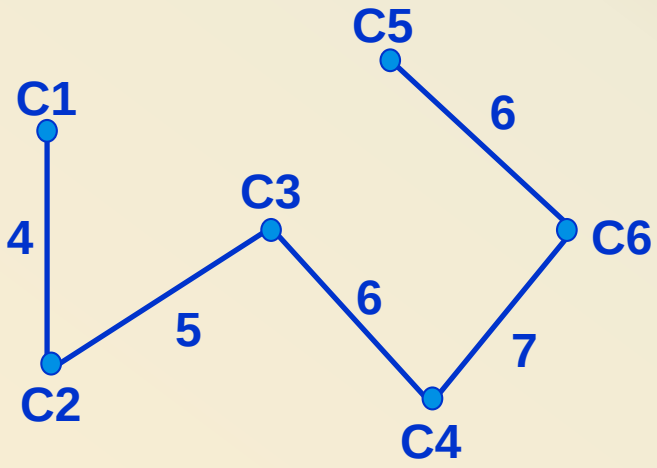



**MST – 1st Solution**

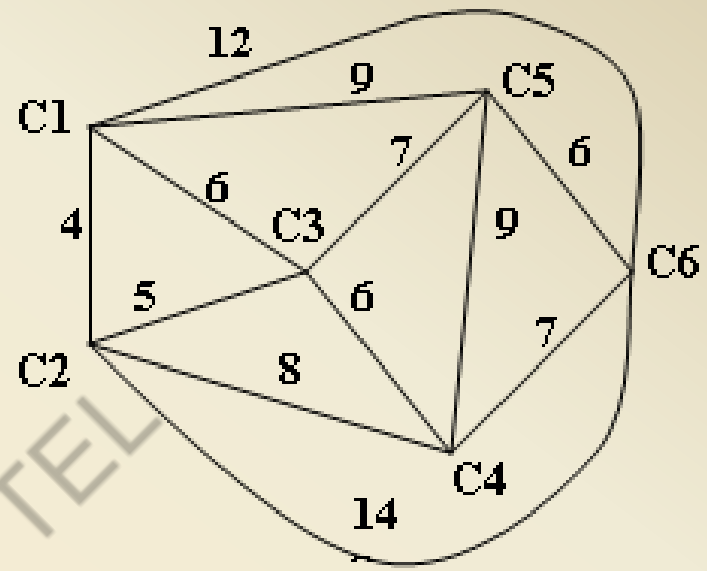



**MST – Alternate Solution**

# Minimal spanning tree

## Prim's algorithm

**Find Minimal Spanning tree (MST) for the graph shown:**




**How do we proceed to obtain MST Using Prim's Algorithm?**

- ***First, we start from a node – say C1***
- ***Now connect C1 with its nearest neighbor – is it C2?***
- ***Now, consider C1 and C2 as one subgraph – connect it to its nearest neighbor (node) – is it C3 through C2-C3?***
- ***Next, take C1, C2, C3 as one subgraph – connect it to C4 – why?***
- ***Now, C1, C2, C3, C4 is the subgraph – connect it to C5 or C6?***
- ***What should be last node?***
- ***Are all the nodes or vertices chosen?***

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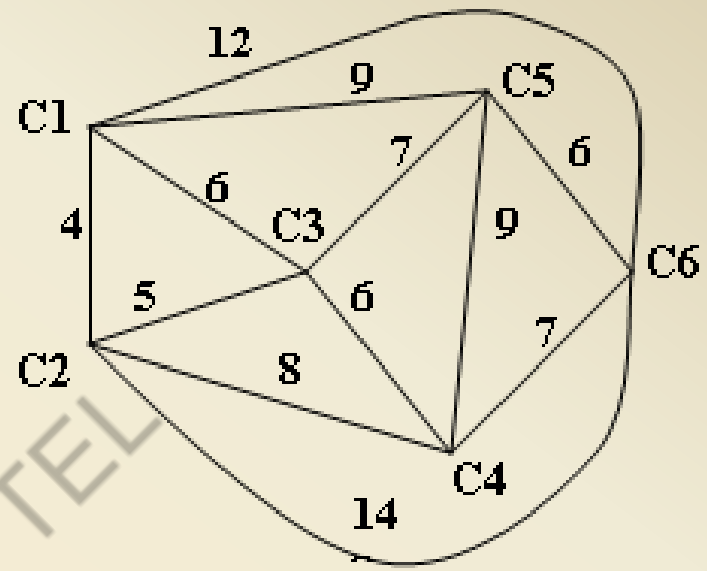
# Minimal spanning tree

**Prim's algorithm**

**Find Minimal Spanning tree (MST) for the graph shown:**




**Two Solutions emerge – both with a total weight of 28.**

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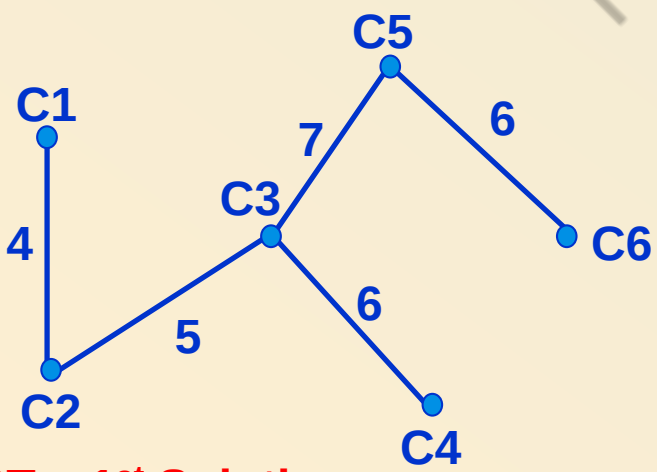
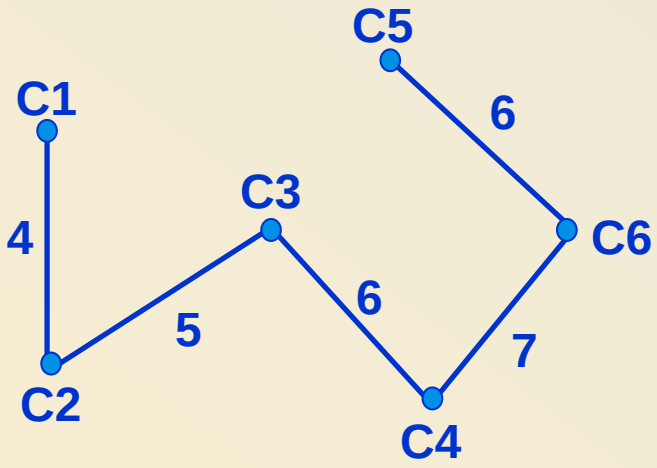


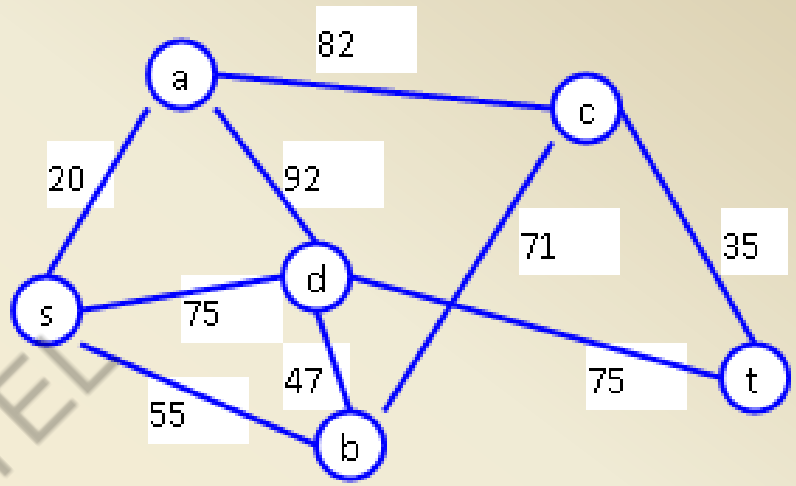

**MST – 1st Solution**




**MST – Alternate Solution**

# MST Exercise

***Find Minimal Spanning tree (MST) for the graph shown using:***




1) ***Kruskal's Algorithm***
2) ***Prim's Algorithm***

***Also, find Maximal Spanning tree.***

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# Cutsets

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# Cutset

- A Cutset is a set of edges in a graph G, whose removal leaves the graph G disconnected provided that no proper subset of these edges does the same.
- A minimal set of edges whose removal leaves graph G disconnected.
- Find cutsets in the graph shown below:

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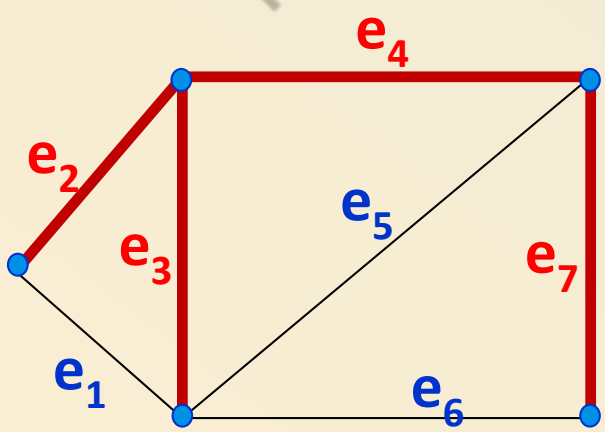
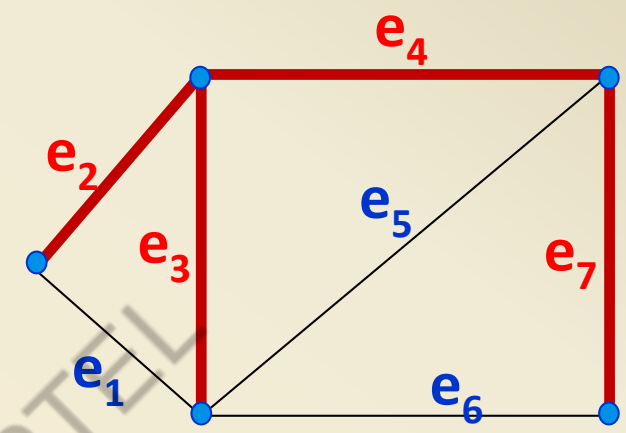


# Cutset

**Cutsets of the Graph:**

- $\{e_1, e_2\}$
- $\{e_1, e_3, e_4\}$
- $\{e_1, e_3, e_5, e_6\}$
- $\{e_1, e_3, e_5, e_7\}$
- $\{e_2, e_3, e_4\}$
- $\{e_2, e_3, e_5, e_6\}$
- $\{e_2, e_3, e_5, e_7\}$
- $\{e_4, e_5, e_6\}$
- $\{e_4, e_5, e_7\}$
- $\{e_6, e_7\}$

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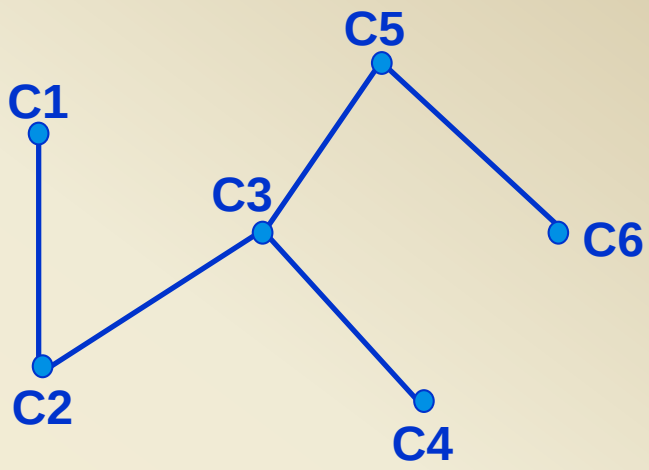
**How to find all the cutsets of a graph?**

# Eccentricity of Vertices

- Distance is a metric. It is:
  - non negative
  - symmetrical
  - triangular inequality holds




- Eccentricity of a vertex v is the distance of the furthest vertex from v in the graph.
- What is the eccentricity of vertex C1?
- What is the eccentricity of vertex C3?

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# Edge and Vertex Connectivity

- **Edge connectivity (E.C.) - number of edges in the smallest cutset of the graph - minimum number of edges whose removal leaves the graph disconnected**
- **Vertex connectivity (V.C.) - Minimum number of vertices whose removal leaves the graph disconnected**
- **Find the E.C. and V.C. of the following two graphs.**

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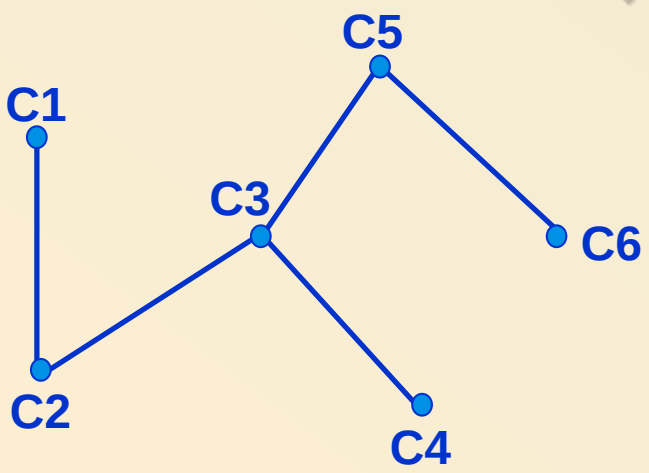
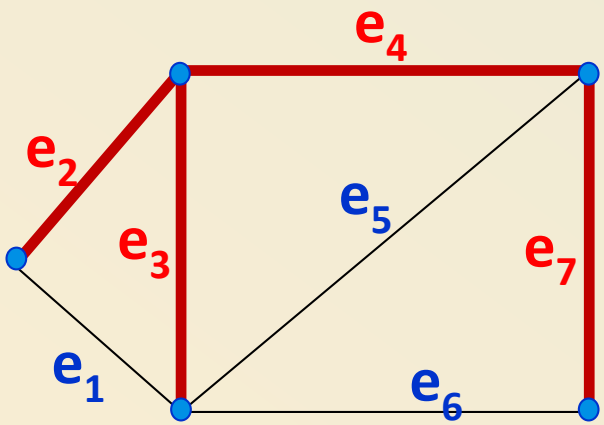
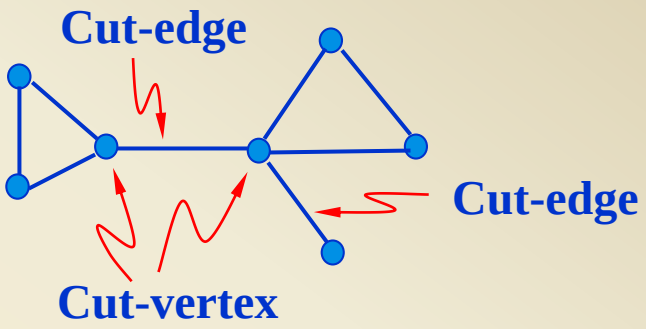


# Cut-Vertex and Cut-Edge

- A **cut-vertex** of a graph is a vertex whose deletion leaves the graph disconnected.
- A **cut-edge** of a graph is an edge whose removal leaves the graph disconnected.

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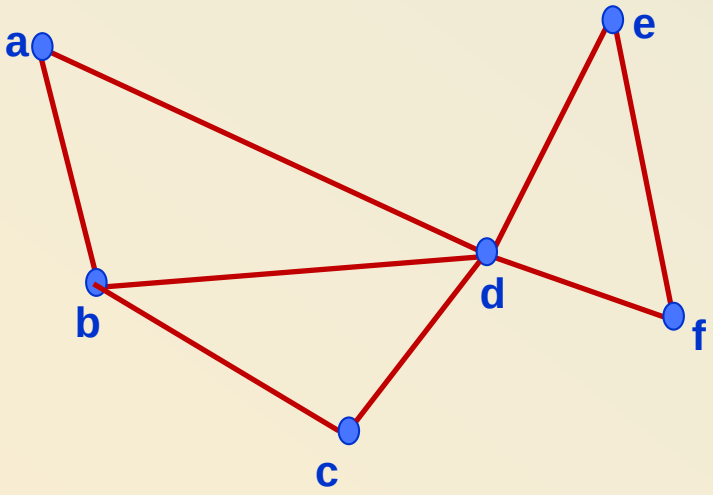
- Is there a **cut-vertex** or **cut-edge** in the graph shown?
- Are there cut-vertex or cut-edge in a **Tree**?



# Relation between E.C. and V.C.

- Edge connectivity (E.C.) - minimum number of edges whose removal leaves the graph disconnected
- Vertex connectivity (V.C.) - Minimum number of vertices whose removal leaves the graph disconnected
- Relation between E.C. and V.C.
- V.C. <= E.C.
- E.C. <= Min $d(v_i)$ $v_i$ in graph G
- V.C. <= Int (2e/n) e: no. of edges; n: no. of vertices
- Question: How will you maximize connectivity in a graph with 6 vertices and 12 edges? What will be its vertex and edge connectivity?

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# THANK YOU

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